Title: Multiclass Disease Classification using Autoencoders and Deep Neural Networks

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Abstract

In this work, we address the challenges of multiclass disease prediction from sparse biomedical data in a clinical diagnosis setting. Using a large dataset with binary symptom indicators and diagnostic labels, we evaluate multiple strategies for addressing extreme class imbalance. Three threshold-based dataset balancing methods are compared against a class-weighted loss approach. For the balanced threshold settings, we develop and test two different deep neural network architectures for the threshold experiments: a baseline classifier and an autoencoder classifier. To address the feature sparsity of the dataset, our analysis pipeline begins by training an autoencoder to extract low-dimensional latent features from the data. These features are then used to train a multi-layer neural network classifier. Model evaluation shows that the autoencoder classifier consistently outperformed the baseline classifier across all three thresholds. Paired t-tests across shared random seeds confirmed the statistically significant improvements. The class-weighted classifier achieved competitive Macro F1 score performance with the baseline and autoencoder classifier at the lowest threshold but failed to produce non-zero recall for certain diseases highlighting the limitations of class-weighted loss methods in large multiclass settings. The results of this project suggest that autoencoder feature compression can help improve the accuracy and consistency of multiclass classification. These machine learning algorithms can be potentially utilized to support doctors in disease predictions based on patient reported symptoms.

Keywords

Autoencoder, Deep Neural Networks, Multiclass Classification, Disease Prediction, Feature Compression, Model Evaluation, Clinical Diagnosis, Healthcare Analytics, Class Weighted Loss, Machine Learning

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